```
import numpy as np
import pandas as pd
import re
from nltk.corpus import stopwords
from nltk.stem.porter import PorterStemmer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score , confusion_matrix , classification_

import nltk
nltk.download('stopwords')
```

```
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
True
```

```
print(stopwords.words('english'))
```

```
['a', 'about', 'above', 'after', 'again', 'against', 'ain', 'all', 'am', 'an', '
```

Data Preprocessing

```
true_df = pd.read_csv('/content/True.csv')
fake_df = pd.read_csv('/content/Fake.csv')
```

```
print(true_df.head())
print(fake_df.head())
```

```
                                               title  \
0  As U.S. budget fight looms, Republicans flip t...
1  U.S. military to accept transgender recruits o...
2  Senior U.S. Republican senator: 'Let Mr. Muell...
3  FBI Russia probe helped by Australian diplomat...
4  Trump wants Postal Service to charge 'much mor...

                                                text       subject  \
0  WASHINGTON (Reuters) - The head of a conservat...  politicsNews
1  WASHINGTON (Reuters) - Transgender people will...  politicsNews
2  WASHINGTON (Reuters) - The special counsel inv...  politicsNews
3  WASHINGTON (Reuters) - Trump campaign adviser ...  politicsNews
4  SEATTLE/WASHINGTON (Reuters) - President Donal...  politicsNews

                 date
0  December 31, 2017
1  December 29, 2017
2  December 31, 2017
3  December 30, 2017
```

```
4  December 29, 2017
                                                title  \
0   Donald Trump Sends Out Embarrassing New Year’...
1   Drunk Bragging Trump Staffer Started Russian ...
2   Sheriff David Clarke Becomes An Internet Joke...
3   Trump Is So Obsessed He Even Has Obama’s Name...
4   Pope Francis Just Called Out Donald Trump Dur...

                                                 text subject  \
0  Donald Trump just couldn t wish all Americans ...    News
1  House Intelligence Committee Chairman Devin Nu...    News
2  On Friday, it was revealed that former Milwauk...    News
3  On Christmas day, Donald Trump announced that ...    News
4  Pope Francis used his annual Christmas Day mes...    News

               date
0  December 31, 2017
1  December 31, 2017
2  December 30, 2017
3  December 29, 2017
4  December 25, 2017
```

```
true_df['label'] = 0
fake_df['label'] = 1
```

1. True News = 0
2. Fake News = 1

```
print(true_df.head())
print(fake_df.head())
```

```
                                                title  \
0  As U.S. budget fight looms, Republicans flip t...
1  U.S. military to accept transgender recruits o...
2  Senior U.S. Republican senator: 'Let Mr. Muell...
3  FBI Russia probe helped by Australian diplomat...
4  Trump wants Postal Service to charge 'much mor...

                                                 text       subject  \
0  WASHINGTON (Reuters) - The head of a conservat...  politicsNews
1  WASHINGTON (Reuters) - Transgender people will...  politicsNews
2  WASHINGTON (Reuters) - The special counsel inv...  politicsNews
3  WASHINGTON (Reuters) - Trump campaign adviser ...  politicsNews
4  SEATTLE/WASHINGTON (Reuters) - President Donal...  politicsNews

                date  label
0  December 31, 2017      0
1  December 29, 2017      0
2  December 31, 2017      0
3  December 30, 2017      0
4  December 29, 2017      0
                                                title  \
0   Donald Trump Sends Out Embarrassing New Year’...
```

```
1   Drunk Bragging Trump Staffer Started Russian ...
2   Sheriff David Clarke Becomes An Internet Joke...
3   Trump Is So Obsessed He Even Has Obama’s Name...
4   Pope Francis Just Called Out Donald Trump Dur...

                                                text subject  \
0  Donald Trump just couldn t wish all Americans ...    News
1  House Intelligence Committee Chairman Devin Nu...    News
2  On Friday, it was revealed that former Milwauk...    News
3  On Christmas day, Donald Trump announced that ...    News
4  Pope Francis used his annual Christmas Day mes...    News

               date  label
0  December 31, 2017      1
1  December 31, 2017      1
2  December 30, 2017      1
3  December 29, 2017      1
4  December 25, 2017      1
```

```
true_df.isnull().sum()
```

	0
title	0
text	0
subject	0
date	0
label	0

dtype: int64

```
fake_df.isnull().sum()
```

	0
title	0
text	0
subject	0
date	0
label	0

dtype: int64

```
print(true_df.shape)
print(fake_df.shape)
```

```
(21417, 5)
(23481, 5)
```

```
print(true_df['date'])
```

```
0        December 31, 2017
1        December 29, 2017
2        December 31, 2017
3        December 30, 2017
4        December 29, 2017
                ...
21412      August 22, 2017
21413      August 22, 2017
21414      August 22, 2017
21415      August 22, 2017
21416      August 22, 2017
Name: date, Length: 21417, dtype: object
```

```
print(true_df.columns)
print(fake_df.columns)
```

```
Index(['title', 'text', 'subject', 'date', 'label'], dtype='object')
Index(['title', 'text', 'subject', 'date', 'label'], dtype='object')
```

```
news_dataset = pd.concat([true_df, fake_df], axis=0)
```

```
news_dataset = news_dataset.sample(frac = 1, random_state=42).reset_index(drop=
```

```
news_dataset["id"] = news_dataset.index
```

```
news_dataset.head()
```

	title	text	subject	date	label	id
0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1	0
1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1	1
2	Mike Pence's New DC Neighbors Are HILARIOUSLY...	Mike Pence is a huge homophobe. He supports ex...	News	December 3, 2016	1	2
3	California AG pledges to defend birth control ...	SAN FRANCISCO (Reuters) - California Attorney ...	politicsNews	October 6, 2017	0	3
4	AZ RANCHERS Living On US-Mexico Border Destroy...	Twisted reasoning is all that comes from Pelos...	politics	Apr 25, 2017	1	4

Next steps: Generate code with news_dataset New interactive sheet

```
cols = ["id"] + [col for col in news_dataset.columns if col != "id"]
news_dataset = news_dataset[cols]
```

```
news_dataset.head()
```

	id	title	text	subject	date	label
0	0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1
1	1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1
2	2	Mike Pence's New DC Neighbors Are HILARIOUSLY...	Mike Pence is a huge homophobe. He supports ex...	News	December 3, 2016	1
3	3	California AG pledges to defend birth control ...	SAN FRANCISCO (Reuters) - California Attorney ...	politicsNews	October 6, 2017	0
4	4	AZ RANCHERS Living On US-Mexico Border Destroy...	Twisted reasoning is all that comes from Pelos...	politics	Apr 25, 2017	1

Next steps: Generate code with news_dataset New interactive sheet

merging the news title and news text

```
news_dataset['content'] = news_dataset['title'] + ' ' + news_dataset['text']
```

```
news_dataset.head()
```

	id	title	text	subject	date	label	content
0	0	BREAKING: GOP Chairman Grassley Has Had Enoug...	Donald Trump s White House is in chaos, and th...	News	July 21, 2017	1	BREAKING: GOP Chairman Grassley Has Had Enoug...
1	1	Failed GOP Candidates Remembered In Hilarious...	Now that Donald Trump is the presumptive GOP n...	News	May 7, 2016	1	Failed GOP Candidates Remembered In Hilarious...
2	2	Mike Pence’s New DC Neighbors Are HILARIOUSLY...	Mike Pence is a huge homophobe. He supports ex...	News	December 3, 2016	1	Mike Pence’s New DC Neighbors Are HILARIOUSLY...
3	3	California AG pledges to defend birth control ...	SAN FRANCISCO (Reuters) - California Attorney ...	politicsNews	October 6, 2017	0	California AG pledges to defend birth control ...
4	4	AZ RANCHERS Living On US-Mexico Border Destroy...	Twisted reasoning is all that comes from Pelos...	politics	Apr 25, 2017	1	AZ RANCHERS Living On US-Mexico Border Destroy...

Next steps: Generate code with `news_dataset` New interactive sheet

```
# Separating the data and label
X = news_dataset.drop(columns='label', axis=1)
Y = news_dataset['label']
```

```
print(X)
print(Y)
```

```
          id                                              title  \
0          0   BREAKING: GOP Chairman Grassley Has Had Enoug...
1          1   Failed GOP Candidates Remembered In Hilarious...
2          2   Mike Pence’s New DC Neighbors Are HILARIOUSLY...
3          3  California AG pledges to defend birth control ...
4          4  AZ RANCHERS Living On US-Mexico Border Destroy...
...      ...                                                ...
44893  44893  Nigeria says U.S. agrees delayed $593 million ...
44894  44894                  Boiler Room #62 – Fatal Illusions
44895  44895  ATHEISTS SUE GOVERNOR OF TEXAS Over Display on...
44896  44896  Republican tax plan would deal financial hit t...
```

```
44897  44897  U.N. refugee commissioner says Australia must ...

                                                   text          subject  \
0      Donald Trump s White House is in chaos, and th...             News
1      Now that Donald Trump is the presumptive GOP n...             News
2      Mike Pence is a huge homophobe. He supports ex...             News
3      SAN FRANCISCO (Reuters) - California Attorney ...     politicsNews
4      Twisted reasoning is all that comes from Pelos...         politics
...                                                  ...              ...
44893  ABUJA (Reuters) - The United States has formal...        worldnews
44894  Tune in to the Alternate Current Radio Network...      Middle-east
44895   I m convinced the Freedom From Religion group...  Government News
44896  WASHINGTON (Reuters) - The Republican tax plan...     politicsNews
44897  SYDNEY (Reuters) - The U.N. High Commissioner ...        worldnews

                     date                                            content
0           July 21, 2017   BREAKING: GOP Chairman Grassley Has Had Enoug...
1             May 7, 2016   Failed GOP Candidates Remembered In Hilarious...
2        December 3, 2016   Mike Pence’s New DC Neighbors Are HILARIOUSLY...
3        October 6, 2017   California AG pledges to defend birth control ...
4            Apr 25, 2017  AZ RANCHERS Living On US-Mexico Border Destroy...
...                   ...                                                ...
44893  December 27, 2017   Nigeria says U.S. agrees delayed $593 million ...
44894       June 29, 2016  Boiler Room #62 – Fatal Illusions Tune in to t...
44895        Feb 27, 2016  ATHEISTS SUE GOVERNOR OF TEXAS Over Display on...
44896   November 2, 2017   Republican tax plan would deal financial hit t...
44897   November 1, 2017   U.N. refugee commissioner says Australia must ...

[44898 rows x 6 columns]
0        1
1        1
2        1
3        0
4        1
        ..
44893    0
44894    1
44895    1
44896    0
44897    0
Name: label, Length: 44898, dtype: int64
```

Stemming:

Stemming is the process of reducing a word to its Root word

```
port_stem = PorterStemmer()
```

```
def stemming(content):
    stemmed_content = re.sub('[^a-zA-Z]',' ', content)
    stemmed_content = stemmed_content.lower()
    stemmed_content = stemmed_content.split()
    stemmed_content = [port_stem.stem(word) for word in stemmed_content if not
```

```
    stemmed_content = ' '.join(stemmed_content)
    return stemmed_content
#
```

```
news_dataset['content'] = news_dataset['content'].apply(stemming)
```

```
print(news_dataset['content'])
```

```
0        break gop chairman grassley enough demand trum...
1        fail gop candid rememb hilari mock eulog video...
2        mike penc new dc neighbor hilari troll homopho...
3        california ag pledg defend birth control insur...
4        az rancher live us mexico border destroy nanci...
                                ...
44893    nigeria say u agre delay million fighter plane...
44894    boiler room fatal illus tune altern current ra...
44895    atheist sue governor texa display capitol grou...
44896    republican tax plan would deal financi hit u u...
44897    u n refuge commission say australia must stop ...
Name: content, Length: 44898, dtype: object
```

```
X = news_dataset['content'].values
Y = news_dataset['label'].values
```

```
print(X)
```

```
['break gop chairman grassley enough demand trump jr testimoni donald trump whit
 'fail gop candid rememb hilari mock eulog video donald trump presumpt gop nomin
 'mike penc new dc neighbor hilari troll homophob bigot mike penc huge homophob
 ...
 'atheist sue governor texa display capitol ground convinc freedom religion grou
 'republican tax plan would deal financi hit u univers washington reuter republi
 'u n refuge commission say australia must stop unfold humanitarian crisi png sy
```

```
print(Y)
```

```
[1 1 1 ... 1 0 0]
```

```
Y.shape
```

```
(44898,)
```

```
# Converting the textual data to numerical data

vectorizer = TfidfVectorizer()
vectorizer.fit(X)

X = vectorizer.transform(X)
```

```
print(X)
```

```
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 6904081 stored elements and shape (44898, 89868)>
  Coords        Values
  (0, 570)      0.06055486453372135
  (0, 2286)     0.04253552689623891
  (0, 2301)     0.025666202113816185
  (0, 3031)     0.063093661763644
  (0, 3388)     0.047259753130124114
  (0, 6574)     0.07041583382179394
  (0, 7853)     0.0898677958851911
  (0, 7886)     0.059913473329118644
  (0, 9454)     0.048097596645320734
  (0, 10324)    0.08231690448262517
  (0, 11174)    0.03296114815015033
  (0, 11631)    0.04919300578107092
  (0, 12522)    0.05138750652591612
  (0, 12642)    0.06680808165549404
  (0, 13496)    0.06811316396970238
  (0, 13713)    0.07112856852592032
  (0, 13991)    0.053000278097355684
  (0, 14098)    0.037972261733148126
  (0, 14836)    0.0869210456407381
  (0, 14928)    0.06842646468851935
  (0, 15149)    0.04165264736324387
  (0, 15153)    0.051876038877662234
  (0, 15333)    0.08494945735141231
  (0, 15663)    0.056535937666043194
  (0, 15765)    0.029199968119368196
  :     :
  (44897, 77773)        0.03757368913834899
  (44897, 77960)        0.01293068706666485
  (44897, 78688)        0.042561076129468856
  (44897, 78704)        0.040544018114292965
  (44897, 79058)        0.05779728742472851
  (44897, 80447)        0.04313047656027923
  (44897, 81481)        0.07874060468713938
  (44897, 81621)        0.04451315198091657
  (44897, 82255)        0.03796982127137953
  (44897, 82341)        0.030435493815230488
  (44897, 82450)        0.03850229932669336
  (44897, 83260)        0.033882067136460456
  (44897, 83611)        0.02756733398300722
  (44897, 84581)        0.020184813480884824
  (44897, 84769)        0.13810212379455322
  (44897, 85054)        0.018516634241257903
  (44897, 85198)        0.01690775168103281
  (44897, 86130)        0.030132938735777468
  (44897, 86169)        0.040827065890862685
  (44897, 86180)        0.059445897055736864
  (44897, 86481)        0.04853978108940374
  (44897, 86551)        0.015240029412780992
  (44897, 86624)        0.029560471446670476
```

```
  (44897, 87936)        0.024069000877548073
  (44897, 88056)        0.021027619607628736
```

Splitting the dataset to training & test data

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size=0.2, strati
```

Applying Logistic Regression Model

```
model = MultinomialNB()
```

```
model.fit(X_train, Y_train)
```

```
▾ MultinomialNB
MultinomialNB()
```

Evaluating accuracy score

```
X_train_prediction = model.predict(X_train)
training_data_accuracy = accuracy_score(X_train_prediction, Y_train)
```

```
print('Accuracy score of the training data : ', training_data_accuracy)
```

```
Accuracy score of the training data :  0.9410601926610613
```

```
X_test_prediction = model.predict(X_test)
test_data_accuracy = accuracy_score(X_test_prediction, Y_test)
```

```
print('Accuracy score of the test data : ', test_data_accuracy)
```

```
Accuracy score of the test data :  0.9366369710467706
```

```
# Confusion Matrix
cm = confusion_matrix(Y_test, X_test_prediction)
print("Confusion Matrix:\n", cm)

# Classification Report
report = classification_report(Y_test, X_test_prediction)
print("\nClassification Report:\n", report)
```

```
Confusion Matrix:
 [[3931  353]
 [ 216 4480]]
```

```
Classification Report:
              precision    recall  f1-score   support

           0       0.95      0.92      0.93      4284
           1       0.93      0.95      0.94      4696

    accuracy                           0.94      8980
   macro avg       0.94      0.94      0.94      8980
weighted avg       0.94      0.94      0.94      8980
```